CREW RESOURCE MANAGEMENT (CRM)

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By Alex Pereslavtsev - http://www.airliners.net/photo/Transaero-Airlines/Boeing-777-212-ER/1984464/L/ , GFDL 1.2, https://commons.wikimedia.org/w/index.php?curid=16594678

TOPICS

- Origins of CRM
  - ICAO SHELL Model [framework]
  - Accidents in the 1970s
- Components of CRM [mitigation]
  - Training clusters
  - Behavioral markers
- Evolution of CRM

ORIGINS OF CRM

INTERNATIONAL CIVIL AVIATION ORGANIZATION (ICAO) SHELL MODEL (EDWARDS, 1972)

- **Software** - the rules/regulations, procedures, written documents etc., which are part of the standard operating procedures. *[Note: Different from "software" today.]*
- **Hardware** - the Air Traffic Control suites, or flight deck controls, their configuration, controls and surfaces, displays and functional systems.
- **Environment** - the situation in which the L-H-S system must function, the social and economic climate as well as the natural environment, both inside and outside
- **Liveware** - the human beings - the controller with other controllers, flight crews, engineers and maintenance personnel, ground crews, management and administration people

https://skybrary.aero/articles/icao-shell-model and
The History of CRM on YouTube

SINCE THE SHELL MODEL…

- Expanded checklists, safety procedures, and regulations
- More complex flight deck technology, complex displays
- HFACS and Swiss Cheese Model
- A recognition that human (liveware) error was particularly important to address, based on a series of accidents in the 1970s

SOME ACCIDENTS THAT LED TO CRM CONCEPT

- 1972 Eastern 401 – crew fixated on burnt out lightbulb for nose gear
- 1977 Tenerife, Canary Islands, Spain
  - Surface collision of two Boeing 747 aircraft; deadliest accident in aviation history with 538 fatalities
  - KLM aircraft took off while a Pan American aircraft was taxiing in opposite direction down the same runway
  - The Captain of the KLM aircraft made several errors that the First Officer and Flight Engineer did not challenge sufficiently; authority gradient.
- 1978 United Airlines Flight 173, Portland, Oregon
  - Ran out of fuel while circling to check landing gear
  - First Officer and Flight Engineer not assertive with dominant captain while running out of fuel. Pilot did not comprehend the fuel situation, turned away from airport with less than five minutes of fuel.
- Other CRM-related accidents:
  - https://www.faa.gov/lessons_learned/transport_airplane/accidents/threat_categories

➢ 1979 Industrial workshop sponsored by NASA. As an outcome, United Airlines started the first Command, Leadership, and Resource Management training (CLR) in 1981

SOME OBSERVED PROBLEMS

- “Chain of command" approach was more the norm
- Inadequate leadership
- Failure to communicate intentions and plans
- Failure to set priorities

WHAT IS CRM?

OVERVIEW

- CRM is a system that uses all available resources to promote safety.
- CRM is concerned with the cognitive and interpersonal skills needed to manage resources within an organized system, not with technical knowledge and skills.
- CRM aims to foster a climate or culture where authority may be respectfully questioned.
- CRM training thus emphasizes communications, situational awareness, problem solving, decision making, and effective teamwork.

FAA Safety Briefing, Nov/Dec 2019 Issue. Online version at https://express.adobe.com/page/Hs9PMKOBdKOAA/

GOALS OF CRM

- Reduce crew error
- Operate more effectively as teams
- Learning, practicing, and getting feedback under low stress should improve performance in emergency/stressful situations.

MORE ABOUT CRM

- CRM is another defense layer, designed to reduce the preconditions for unsafe acts
  - Use an open communication process
  - Prioritizing
  - Workload management
- CRM is about managing resources (including people, systems, etc.) while meeting time demands and safety goals.
- Technical proficiency is a separate prerequisite. Goal is to blend human and technical skills.
- Applicable to all types of operations including military, and even single-pilot operations.
- Initial and recurrent training is important, plus continuous reinforcement (practice) and feedback.
- CRM instruction is also given to medical teams, nuclear reactor control teams, and emergency response teams.

CRM FOR NON-EMERGENCY SITUATIONS

There are many normal high-workload situations that involve managing many tasks, e.g.,

- Planning for a weather diversion
  - Requires extra communication and weather assessment tasks
  - Need to effective use all available resources
- Handling in-flight route amendments
  - Deciding whether to accept the revised route
  - Prioritizing tasks related to route amendments, especially with complex and precise routing and timing requirements
- Managing flight modes
  - Brief/communicate any plans or changes in use of automated systems to crew members

❖ *Excellent crew coordination is needed on every flight!*

TRAINING CLUSTERS

- **Communication processes & Decision making**
  - Briefings and debriefings
    - e.g., establish an environment for open communication, establish a team concept, identify potential problems
  - Inquiries, advocacy, and assertions
    - e.g., speak up and state information with appropriate persistence until resolution, questions are encouraged, seek help fromothers
  - Self-critique (decisions and actions)
    - e.g., identify both positive and negative aspects of crew performance, feedback should be specific, objective, and constructive
  - Conflict resolution
- **Team building and maintenance**
  - Leadership and followership
  - Operational concerns
  - interpersonal climate
- **Workload management and situational awareness**
  - Preparation and planning
    - e.g., demonstrate and express a mental model of the situation, share it with others
  - Vigilance
  - Workload distribution and avoiding distractions

See Appendix 1 of CRM AC for more examples

OBSERVABLE BEHAVIORS

Table 7: Identified CRM skills exhibited by a competent team

CRM Skill Identified	Definition
Communication (also known as Closed-loop communication)	Ability of two or more team members to clearly and accurately send and receive information or commands and to provide useful feedback.
Briefing (Mission analysis, Planning)	Ability of team members to develop plans of action by organising team resources, activities and responses to ensure tasks are completed in an integrated and synchronised manner.
Backup behaviour (Advocacy)	Ability of team members to anticipate the needs of others through accurate knowledge about each other's responsibilities, including the ability to shift workload between members to create balance during periods of high workload or pressure.
Mutual performance monitoring (Workload management)	Ability of a team member to accurately monitor other team members' performance, including giving, seeking and receiving task-clarifying feedback.
Team leadership (Management)	Ability of a team leader to direct and coordinate the activities of team members, encourage team members to work together; assess performance; assign tasks; develop team knowledge, skills and abilities; motivate; plan and organise; and establish a positive team atmosphere.
Decision making (judgment, problem solving)	Ability of team members to gather and integrate information, make logical and sound judgments, identify alternatives, consider the consequences of each alternative, and select the best one.

CRM Skill Identified	Definition
Task-related assertiveness (confidence, aggressiveness, authoritarian)	Willingness/readiness of team members to communicate their ideas, opinions and observations in a way that is persuasive to other team members and to maintain a position until convinced by the facts that other options are better.
Team adaptability (flexibility)	Ability of team members to alter a course of action or adjust strategies when new information becomes available.
Shared situation awareness (shared mental models, situation assessment)	Ability of team members to gather and use information to develop a common understanding of the task and team environment.

Salas, E., Prince, C., Bowers, C., Stout, R., Oser, R.L., & Cannon-Bowers, J.A. (1999). A methodology for enhancing crew resource management training. *Human Factors, 41,* p.163. (Referenced in QF32 accident report.)

EVOLUTION OF CRM

CRM IMPLEMENTATION HAS EVOLVED

- Expanded the concept of "crew"
  - Pilots on the flightdeck (captain/first officer), then included the cabin crew, maintenance, and other parts of organization, e.g., dispatchers and even air traffic controllers.
- Teaching format – integrated with flight training rather than separate module
- Now seen as error management-because error is inevitable
- Non-punitive view of error
- *Four "generations" over the first 20 years*

FIRST GENERATION OF CRM

- Focus on psychological testing and general concepts and strategies (e.g., for leadership), but didn’t provide clear definitions of appropriate behavior.
- Design was based on management training, with games and exercise unrelated to aviation.
- Some programs included line-oriented flight training in a flight simulator for practice without jeopardy.
- Overall acceptance, but some resistance (“charm school”)

SECOND GENERATION OF CRM

- Second NASA workshop in 1986
- Goal to embed CRM training in flight training, not to keep it as a separate course
- Changed name from “cockpit” resource management to “crew” resource management
- More team-oriented
  - Briefing strategies, situation awareness and stress management, decision-making strategies
- Greater acceptance, but still criticized for using “psycho-babble”

THIRD GENERATION OF CRM

- Effort to integrate CRM with technical training
- Began to include organizational culture in the training, recognizing that it affects CRM
- Added modules around the use of flight deck automated systems and CRM
- Advanced training given to pilots who check/evaluate other pilots
- Developed separate CRM programs for flight attendants, dispatchers, and maintenance personnel
- Unfortunate unintended consequence was a reduced focus on reduction of human error

FOURTH GENERATION OF CRM

- Advanced Qualification Program (AQP) introduced by FAA in 1990.
- Incorporated CRM concepts into checklists and standard operating procedures
- CRM fully integrated with flight training

FAA ADVANCED QUALIFICATION PROGRAM (AQP) PRINCIPLES

- Participation is voluntary
- Employs innovative training and qualification concepts with the regulatory flexibility to tailor training to individual company circumstances
- Uses performance data to drive curriculum changes
- Qualification is based on individual and team performance, using progressive evaluations of proficiency objectives, and the structure and maintenance of all elements (curriculum, facilities, training equipment, instructors, evaluators, courseware, and quality assurance) of the program
- Data collection and analysis to validate individual and team proficiency, and the AQP itself
- Training is systematically developed with an audit trail for all training and data requirements

https://www.faa.gov/training_testing/training/aqp

WHO USES AQP

- 90% of Large 121 carriers (over 1000 pilots) utilize AQP
- 100% of Medium size 121 carriers(501-999 pilots)
- 5% of Small size 121 carriers
- Over 90% of U.S. airline pilots train under AQP

CRITIQUE OF FOURTH GEN CRM

- Positive outcomes from CRM training were validated.

However—

- Expanding the concept to the other groups of personnel may dilute its effectiveness in reducing the frequency and severity of crew-based errors
- Proceduralizing behaviors might also obscure the value of CRM
- CRM does not reach everyone; some people are resistant
- Acceptance of concepts can decay over time, even with recurrent training
- CRM did not always export well to non-USA cultures

CRM TODAY?

- CRM is now viewed as a part of the organization's "safety culture"
  - Downside--CRM is less salient to crews
- Challenges
  - Automated systems are even more complex to manage
  - Workforce is diversifying
  - In-person training "footprint" is expensive and limited; more training is done individually and virtually
  - Air traffic volume increased; faster tempo of operations (e.g., radio congestion)
  - More precise flight paths (e.g., 4-D trajectories)
  - Society has changed; younger generation is less comfortable interacting in person
- Also see online article from Flight Safety Foundation:

https://flightsafety.org/asw-article/what-happened-to-crew-resource-management/

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- https://www.faa.gov/training_testing/training/aqp/library
- https://www.faa.gov/sites/faa.gov/files/2022-11/crmhistory.pdf
- The History of CRM on YouTube

EXTRAS

CRM AS A TOOL FOR ERROR MANAGEMENT

- Human error is inevitable. Try to prevent it, but be prepared to trap errors and mitigate them.
- In 1997, FAA introduced the Aviation Safety Action Program (ASAP) to encourage confidential, non-punitive reporting of errors internal to organizations (e.g., airlines).

➢ We'll talk more about other methods of tracking and analyzing human performance in organizations